

Species and lineage

Region	H37Rv region	Usually deleted in	Segments found	Depth vs control	Result
RD1	4,350,251-4,359,740	BCG, Dassie bacillus	20 of 20	0.80	+ present
RD4	1,696,001-1,708,740	M. bovis, BCG (some M. canettii)	0 of 20	0.00	– deleted
RD7	2,208,001-2,220,740	M. africanum lineage 6, animal lineages	0 of 20	0.00	– deleted
RD9	2,330,051-2,332,140	M. africanum (lineages 5, 6), animal lineages	0 of 8	0.00	– deleted
RD12	3,485,101-3,487,540	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)	0 of 20	0.00	– deleted

Lineage SNP	Position (H37Rv)	Gene	Reads with lineage allele	Other reads
BOV_AFRI	1,882,180	Rv1662	62	0
BOV	2,831,482	Rv2515c	56	0

Spacer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Count	56	47	0	58	63	0	76	0	0	0	0	0	50	43	38	0	36	29	45	50	47	51
Spacer	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	
Count	50	54	140	74	73	78	57	56	50	46	39	41	45	45	54	53	0	0	0	0	0	

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RD1	4,350,251-4,359,740	BCG, Dassie bacillus	20 of 20	1.00	+ present
RD4	1,696,001-1,708,740	M. bovis, BCG (some M. canettii)	0 of 20	0.00	– deleted
RD7	2,208,001-2,220,740	M. africanum lineage 6, animal lineages	0 of 20	0.00	– deleted
RD9	2,330,051-2,332,140	M. africanum (lineages 5, 6), animal lineages	0 of 8	0.00	– deleted
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Lineage SNP	Position (H37Rv)	Gene	Reads with lineage allele	Other reads
BOV_AFRI	1,882,180	Rv1662	1	0
BOV	2,831,482	Rv2515c	1	0

Spacer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Count	1	1	0	1	1	0	1	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1
Spacer	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	
Count	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	

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RD1	4,350,251-4,359,740	BCG, Dassie bacillus	20 of 20	1.00	+ present
RD4	1,696,001-1,708,740	M. bovis, BCG (some M. canettii)	20 of 20	1.00	+ present
RD7	2,208,001-2,220,740	M. africanum lineage 6, animal lineages	20 of 20	1.00	+ present
RD9	2,330,051-2,332,140	M. africanum (lineages 5, 6), animal lineages	8 of 8	1.00	+ present
RD12	3,485,101-3,487,540	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)	20 of 20	1.00	+ present

Lineage SNP	Position (H37Rv)	Gene	Reads with lineage allele	Other reads
4	931,123	Rv0835	1	0
4.9	1,759,252	Rv1552	1	0

Spacer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Count	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	1
Spacer	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	
Count	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1	1	1		

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Definitions and methods

Spoligotype	The presence (1, filled square) or absence (0, empty square) of the 43 spacers of the direct repeat (DR) locus, written as a 43-digit binary pattern, a 15-digit octal code (Dale et al. 2001) or a hexadecimal code. These codes are universal: they identify the pattern itself, and are the ones to use to exchange spoligotypes.
SB number	The name of a pattern in the Mbovis.org database of spoligotypes of the RD9-deleted lineages (M. bovis and other animal-adapted lineages). "Not in Mbovis.org" only means that this database has no name for the pattern, as for all M. tuberculosis patterns: it does not mean that the spoligotype is invalid or new.
SIT	Shared international type of the SITVIT2 database (Institut Pasteur de Guadeloupe), from the SITVIT2 patterns published with SpolLineages (9,656 patterns, 3,850 SITs, 2022 list). "Orphan": a SITVIT2 pattern without SIT. "Not in SITVIT2 list": the pattern is not in this list, which does not include the SITs created since 2022. SITVIT2 lacks many patterns of the animal-adapted lineages: only 603 of the 1,976 Mbovis.org patterns have a SIT in this list. For these lineages, the SB number is the reference name, and no closest SIT is given for a pattern with an SB number.
Spoligotype family	SITVIT2 family of the pattern (e.g. Beijing, LAM3, BOV_1), and the spoligotype families typical of the lineage (Coll et al. 2014). Families are labels of patterns, not phylogenetic lineages.
Lineage	From the 62 SNPs of the barcode of Coll et al. (2014): the reads carrying each allele are counted with exact 31-mers; a lineage is called when at least 80% of the reads (at least 3, or 1 contig) carry its allele.

Regions of difference

Regions of difference (RD) are determined *in silico*, from the sequencing data, without PCR. Each region is the part of the H37Rv genome missing from M. bovis AF2122/97 (RD4, RD7, RD9, RD12) or from BCG Pasteur (RD1), and is represented by 100 bp segments inside it (table below), chosen to be found in all the MTBC genomes that have the region and in none of those lacking it or of six non-tuberculous mycobacteria. A segment is found when Seal (25-mers, 1 mismatch) finds it in the reads at a depth of at least 5% of the depth of MTBC-specific control regions, or in the assembly. A region is present when all its segments are found (in reads, up to 10% of them, at least 1, may be missing: low depth); deleted when at most 10% of its segments (at least 1: stray reads) are found; partially deleted in between, with the H37Rv coordinates of the missing segments; present at reduced depth when, in reads, its segments are found at less than 50% of the control depth (mixed sample?).

Region	H37Rv region (NC_000962.3)	Segments	Usually deleted in
RD1	4,350,251-4,359,740 (9,490 bp)	20	BCG, Dassie bacillus
RD4	1,696,001-1,708,740 (12,740 bp)	20	M. bovis, BCG (some M. canettii)
RD7	2,208,001-2,220,740 (12,740 bp)	20	M. africanum lineage 6, animal lineages
RD9	2,330,051-2,332,140 (2,090 bp)	8	M. africanum (lineages 5, 6), animal lineages
RD12	3,485,101-3,487,540 (2,440 bp)	20	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)

The + and – of the RD profile therefore mean that the DNA of the region is present or absent, like a PCR with primers inside the region (amplification = present). spoligotyper does not measure amplicon sizes or deletion junctions: assays that distinguish RDs by the size of an amplicon spanning the region may report partial or strain-specific deletions differently. A partially deleted region counts as + when at least half of its segments are found: for example, M. microti lacks the part of RD1 inside its own RD1mic deletion (reported as partially deleted) and is RD1 + in the RD profile, as in the classical RD PCR scheme. The species is read from the RD profile, refined with the lineage SNPs and the spacers.

Spacers and codes

Spacers are counted with Seal (BBTools): each of the 43 spacers is searched as a single 25-mer on both strands, allowing 1 mismatch (k=25, hdist=1, rcomp=t, maskmiddle=f, ambiguous=all). A spacer is present when it is found in at least the minimum count of reads (contigs for assemblies). The fraction of MTBC reads is the depth of the control regions divided by the depth expected from the number of bases.

Run information

Operator	Example
User	example
Computer	example
Operating system	Linux-6.8.0-139-generic-x86_64-with-glibc2.39
Started	2026-09-28 16:27:09 EDT
Finished	2026-09-28 16:27:21 EDT
Duration	12.0 s
Working directory	/example
Command	spoligotyper -i data/ -o results/ -t 8 --operator Example

Parameters

Input	/example/data
Output folder	/example/results
Minimum count	default (5 for fastq, 1 for fasta)
Threads	8
Jobs	1
Seal memory	1g
Species and lineage	yes

Software

spoligotyper	0.7.0
Python	3.14.7 (/opt/conda/bin/python3.14)
Seal	/opt/conda/bin/seal.sh
BBTools	40.02
Java	openjdk version "25.0.2-internal" 2026-01-20

Reference data

Spoligotype database	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spoligotype_db.txt 1,976 patterns · MD5 0a88f7b89f746de2a11e1b4af0151e36
Spacer sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spoligo_spacers.fasta 43 spacers · MD5 033c08e66aaa90193973f8d542e94c1
Species markers	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/markers.fasta MD5 87135cd99f26936911997867c6f1964b
Lineage SNP barcode	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/lineage_barcode.tsv MD5 f676fb8f470889ae02cb2f2b6c5970d7
Lineage SNP sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/lineage_snps.fasta MD5 7e662a9baee450e79065705eb7b79021
SIT database	/opt/spoligotyper/sit_database.tsv 9,656 patterns, 3,850 SITs · SHA-256 6ab06a17d0e39929f7c8a383ee8b406ea958fefecea1a9c04bd599208f44f763 SIT database for spoligotyper, converted from SpolLineages Spoligo_list.csv (SITVIT2 patterns)

References

- Kamerbeek J et al. Simultaneous detection and strain differentiation of Mycobacterium tuberculosis for diagnosis and epidemiology. J Clin Microbiol 35:907-914 (1997). doi:10.1128/jcm.35.4.907-914.1997
- Dale JW et al. Spacer oligonucleotide typing of bacteria of the Mycobacterium tuberculosis complex: recommendations for standardised nomenclature. Int J Tuberc Lung Dis 5:216-219 (2001).
- Smith NH, Upton P. Naming spoligotype patterns for the RD9-deleted lineage of the Mycobacterium tuberculosis complex; www.Mbovis.org. Infect Genet Evol 12:873-876 (2012). doi:10.1016/j.meegid.2011.08.002
- Brosch R et al. A new evolutionary scenario for the Mycobacterium tuberculosis complex. Proc Natl Acad Sci USA 99:3684-3689 (2002). doi:10.1073/pnas.052548299

Couvin D, Segretier W, Stattner E, Rastogi N. Novel methods included in SpolLineages tool for fast and precise prediction of Mycobacterium tuberculosis complex spoligotype families. Database (Oxford) 2020:baaa108. doi:10.1093/database/baaa108 (SIT database, from SITVIT2)

Coll F et al. A robust SNP barcode for typing Mycobacterium tuberculosis complex strains. Nat Commun 5:4812 (2014). doi:10.1038/ncomms5812

Bushnell B. BBTools. <https://sourceforge.net/projects/bbmap/>

Duceppe M-O. spoligotyper 0.7.0: in silico spoligotyping of Mycobacterium tuberculosis complex genomes. Zenodo. <https://doi.org/10.5281/zenodo.22926160>